A Review on Swarm Resilience Enhancement: Methods and Applications

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Abstract— **Unmanned aerial vehicle (UAV) swarms are increasingly deployed in complex and unpredictable environments due to their scalability, redundancy, and flexibility. However, their performance can be significantly degraded by internal and external disruptions, including communication failures, agent malfunctions, environmental disturbances, and collisions. This paper provides a comprehensive review of resilience enhancement strategies in UAV swarms. We categorize disruptive factors into internal (e.g., communication, coordination, consensus) and external (e.g., collisions, deadlocks, wind, and obstacles), and analyze decentralized algorithms that strengthen resilience. These include the SHARKS protocol, Dec-MDPs, distributed SLAM, Buffered Voronoi Cells (BVC), and deadlock recovery methods. By integrating algorithmic solutions with biological inspirations, we identify promising pathways toward more robust and adaptive swarm systems in real-world applications.**

Keywords— UAV swarms, swarm resilience, decentralized control, collision avoidance, deadlock recovery, distributed SLAM

I. Swarm Resilience

Unmanned aerial vehicles (UAVs) are aircraft that operate without on-board pilots, offering significant potential in domains including surveillance, disaster relief, environmental monitoring, and search and rescue missions [1] [2] [3]. These systems are inherently failure-tolerant and decentralized, enabling them to solve complex tasks that may be challenging for human operators. Drawing inspiration from biological swarm behaviors, such as those observed in ant colonies, bee swarms, and bird flocks, UAV swarms involve the collective behavior of multiple autonomous agents [4]. Compared to individual UAVs, UAV swarms exhibit enhanced and efficient cooperation, allowing them to address intricate problems and perform tasks, including emergent victim assistance in disaster scenarios and dynamic task transitions in emergency response operations [5]. Although UAV swarms offer numerous advantages, their performance can be significantly impacted by various disruptive events, including adverse weather conditions, security threats, loss of communication, collisions with other agents or physical obstacles, navigation challenges in complex environments, and enemy invasion [6] [7]. The loss of a single agent can lead to system-wide failure due to the interdependent nature of data transmission, while a malfunctioning UAV can introduce erroneous sensor data and trigger cascading failures [8]. Consequently, UAV swarms need the capability to maintain operational integrity in the face of such disruptions. This paper categorizes the disruptive events that influence UAV swarms into internal and external factors and explores strategies that enhance swarm resilience.

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II. Swarm Resilience to Internal Factors

Internal factors within UAV swarms, such as communication, coordination, and consensus, play a critical role in maintaining the robustness of these UAV systems. Effective management of these factors is vital to ensure swarm resilience, especially in dynamic environments.

A. Resilience to Communication

For UAV swarms to function effectively, it is essential that individual UAVs maintain reliable communication with each other and with ground control [9] [10]. Effective communication facilitates coordinated actions, ensuring that the swarm operates coherently despite environmental and operational challenges. However, disruptions, such as loss of network connectivity, data transmission failures, and interference between network nodes, can undermine communication channels, potentially leading to mission failure [11].

Decentralization has emerged as one of the most effective approaches in enhancing swarm resilience for communication. Unlike centralized swarms, which rely on a leader, decentralized swarms are failure-tolerant and make decisions based on interactions with neighboring agents. By distributing control and decision-making processes across all agents rather than relying on a central agent, decentralized swarms can carry out their mission despite individual UAV communicative interruptions [12] [13].

Numerous innovative decentralized algorithms have been designed to enhance swarm resilience by enabling individual agents to make decisions locally, resulting in coordinated global behavior. For example, the SHARKS protocol (Secure, Heterogeneous, Autonomous, and Rotational Knowledge for Swarms) mitigates the risk of a central point of failure by allowing individual agents to encircle a target without relying on centralized control [14]. The swarm is allowed rotational movement around the target while individual agents distance themselves from other agents. Each agent relies solely on local information, such as the distance from its nearest neighbor and target, to maintain a specific position in the swarm formation. Global cohesion is achieved through the rotational movement around the target of the swarm, while individual agents adjust to maintain spacing from others. This ensures that even without centralized oversight, the swarm can reorganize based on local information while also maintaining global cohesion. Furthermore, decentralized

Markov Decision Processes (Dec-MDPs) provide a mathematical model that allows each UAV to make its own decision based on local information, without the need for centralized control [15]. This optimizes decision-making at the individual level and ensures cooperation in unpredictable environments. Furthermore, Decentralized Collaborative Visual-Inertial Simultaneous Localization and Mapping (SLAM) enables UAVs to work together to construct and share maps of their environment in real time [16], further contributing to resilience in communication disruptions by ensuring environmental constraints are accounted for.

B. Resilience to Coordination and Consensus

Coordination and consensus are critical for UAV swarms to maintain formation and operate collaboratively. Consensus enables the swarm to make decisions collectively, despite potential differences in individual UAVs' local observations. Thus, enhancing resilience in coordination and consensus is crucial in the presence of disruptive events.

In communication, decentralization improves resilience to consensus by distributing decision making across the swarm. The Consensus-Based Bundle Algorithm (CBBA) [17] is one example of a decentralized approach that allows flexible task allocation without centralized oversight. Another example is the weighted mean subsequence reduction algorithm (W-MSR) [18], which provides resilience against faulty or compromised nodes, preventing malicious data from negatively impacting the swarm's collective decision-making process. Individual UAV failure can anchor an entire swarm due to incorrect position translation. In such cases, implementing a real-time anomaly detection approach [19] can identify malfunctioning UAVs, allowing the swarm to resume operation with minimal disruption.

III. Swarm resilience to external factors

External factors, including weather conditions, environmental obstacles, and collisions, significantly impact the performance and resilience of UAV swarms. The ability of UAV swarms to adapt to these factors is crucial for their deployment in both civilian and military applications. This section discusses the resilience of UAV swarms to external factors, focusing on collision avoidance and deadlock management, as well as environmental conditions.

A. Buffered Voronoi Cell (BVC) Method of Collision Avoidance

Collisions in UAV swarms are caused by a variety of external factors, including unpredictable environmental disturbances and high-density swarm configurations. Environmental factors such as wind gusts and moving obstacles can lead to unexpected trajectory deviations, increasing the risk of mid-air collisions [20] [21]. Moreover, the risk of collision and conflict increases in high-density swarms due to limited maneuvering space and overlapping flight trajectories [22].

To mitigate these challenges, The Buffered Voronoi Cell (BVC) method provides a decentralized collision-avoidance strategy for dynamic vehicles, such as UAVs, moving arbitrarily in space [23] [24]. It is employed in the Crazyflie 2.0 platform and relies exclusively on position information from neighboring robots. This decentralized approach avoids reliance on central coordination, enabling greater robustness in swarm operations under real-world conditions. The method operates through onboard computation using the built-in function "enableCollisionAvoidance(*others*, *ellipsoidradii*)", where the "*others*" parameter contains a list of Crazyflie objects in the swarm and the "*ellipsoidradii*" parameter defines the radii of the collision avoidance ellipsoids representing the shape and volume of the objects used for collision avoidance [25]. The Crazyflie computes its Voronoi cell and navigates toward the point within its cell nearest its goal position. This process is repeated until all robots reach their respective goal positions. Collision avoidance is assured as long as each robot remains within its designated Voronoi cell, which is ensured through a safety radius, preventing overlap between two neighboring UAVs and mitigating potential collisions.

B. The Deadlock Problem in High-Density Swarms

A major limitation of decentralization in high-density swarms is deadlock, occurring when multiple UAVs block each other's paths, causing one or more robots to become unable to reach their target positions [26] [27]. The issue becomes problematic in as swarm density increases, raising the likelihood of robots crossing paths [28]. A related problem is livelock, where a robot oscillates between deadlock and deadlock-recovery states without advancement towards its goal [29].

In practical applications, the right-hand rule (RHR) heuristic used in the geometric algorithm works well for smaller swarms but falters in high-density environments [30]. For instance, in a simulation of 50 robots using Matplotlib, deadlock occurred for approximately 6.4 seconds, whereas a simulation of 100 robots failed to make any progress at all.

The Deadlock Prediction and Recovery Algorithm [31], a decentralized collision avoidance algorithm for dynamic UAV swarms using BVCs, mitigates such an effect. Unlike traditional methods that passively detect deadlocks as they occur, the proposed algorithm employs a proactive three-step framework to anticipate deadlocks before they develop. The first step, deadlock prediction, enables early detection of potential deadlock situations, increasing safety distances and allowing UAVs to maneuver towards less congested paths. The second step, deadlock recovery, facilitates the correction of deadlock. Finally, deadlock recovery success prediction determines whether the system has successfully exited a deadlock state and thus concludes the method.

C. Environmental Factors Impacting UAV Swarm Resilience

Environmental disturbances, notably wind and in-path obstacles, significantly affect the resilience of UAV swarms. Because simulations don't involve the risk of real-world damages, accurately modeling these disruptions in a simulated environment is crucial in preparing UAV systems to face real-world challenges and adapt [32] [33] [34].

Wind disturbances can cause UAVs to deviate from their planned trajectory, potentially affecting communication and task completion. By incorporating random wind vectors into the simulation environment and taking into account factors such as wind speed, direction, and altitude, researchers are able to develop more robust UAV swarms capable of adapting to real-world challenges.

Physical obstacles, such as buildings and trees, present another major challenge for UAV swarms by hindering inter-UAV communication and obstructing movement. Additionally, dynamic obstacles, including birds and moving drones, necessitate real-time trajectory updates to maintain cohesion and mission efficiency. By integrating various types of obstacles into simulation environments, researchers can assess the robustness, adaptability, and reliability of UAV swarm algorithms under challenging conditions. This approach is fundamental to advancing efficient UAV swarm technologies in real-world applications.

IV. Conclusion and Future Work

In this paper, we explored the key factors that contribute to resilience in UAV swarms, focusing specifically on internal and external factors. Through a comprehensive review of various methodologies, UAV swarms have demonstrated their ability to maintain functionality despite facing challenges such as communication failures, agent malfunctions, and environmental concerns. We have highlighted several resilience-enhancing strategies, including the decentralized protocols SHARKS, Dec-MDPs, and SLAM, as well as the BVC method of collision avoidance and the Deadlock Prediction and Recovery Algorithm. Furthermore, we explored the importance of effective coordination and consensus mechanisms such as CBBA and W-MSR.

While the reviewed techniques provide promising swarm resilience enhancement, more emphasis needs to be placed on high-density swarm operations. Future work should focus on advancing the robustness of UAV swarms in such high-density conditions, with particular attention to improving deadlock avoidance and recovery in larger swarms.

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